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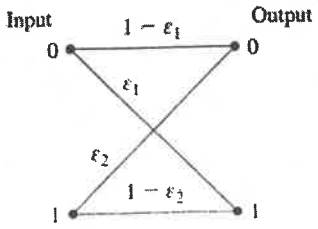
SIXTH SEMESTER (B. Tech), ECE
MID-SEMESTER EXAMINATION, September, 2026

Course Code- ECECC06
Course Title- Probability Theory and Random Process

Time- 1.5 Hours

Max. Marks- 15

Note: - Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	A random experiment has sample space S = {1, 2, 3, 4} with probabilities $p_1=1/2$, $p_2=1/4$, $p_3=1/8$, $p_4=1/8$ (a) Describe how this random experiment can be simulated using tosses of a fair coin. (b) Describe how this random experiment can be simulated using an urn experiment. (c) Describe how this experiment can be simulated using a deck of 52 distinct cards.	1.5	CO1
1b	A binary communication system transmits a signal X that is either a +2 voltage signal or a -2 voltage signal. A malicious channel reduces the magnitude of the received signal by the number of heads it counts in two tosses of a coin. Let Y be the resulting signal. (a) Find the sample space. (b) Find the set of outcomes corresponding to the event "transmitted signal was definitely +2. (c) Describe in words the event corresponding to the outcome Y=0.	1.5	CO1
2a	The lifetime of a device behaves according to the probability law $P[(t, \infty)] = 1/t$ for $t > 1$. Let A be the event "lifetime is greater than 4," and B the event "lifetime is greater than 8." (a) Find the probability of and $A \cap B$, and $A \cup B$. (b) Find the probability of the event "lifetime is greater than 6 but less than or equal to 12."	1.5	CO1
2b	A die is tossed twice and the number of dots facing up is counted and noted in the order of occurrence. Let A be the event "number of dots in first toss is not less than number of dots in second toss," and let B be the event "number of dots in first toss is 6." Find and P[A|B] and P[B|A].	1.5	CO1, CO2
3a	A nonsymmetric binary communications channel is shown in figure. Assume the input is "0" with probability **p** and "1" with probability (**1-p**). (a) Find the probability that the output is 0. (b) Find the probability that the input was 0 given that the output is 1. Find the probability that the input is 1 given that the output is 1. Which input is more probable? 	1.5	CO1
3b	Let X be a binomial random variable that results from the performance of n Bernoulli trials with probability of success p. (a) Suppose that X=1. Find the probability that the single event occurred in the kth Bernoulli trial.	1.5	CO2

	(b) Suppose that X=2. Find the probability that the two events occurred in the jth and k^{th} Bernoulli trials where $j < k$.		
4a	i. Let $\Phi_X(\omega)$ be the characteristic function of an exponential random variable. What random variable $\Phi_X^n(\omega)$ does correspond to? ii. Let **X** be the number of successes in **n** Bernoulli trials where the probability of success is p. Let **Y=X/n** be the average number of successes per trial. Apply the Chebyshev inequality to the event **{|Y-p|>a|}**.	**1.5**	**CO2**
4b	A random variable X has an exponential distribution with probability distribution function is given by $f(x) = 3e^{-3x}$; for x>0 = 0 otherwise. Find probability that X is not less than 2.	**1.5**	**CO3**
5a	Let X be the difference and let Y be the sum of the number of heads obtained when Rashmi and Deepa each flip a fair coin twice. (a) Describe the underlying space S of this random experiment and show the mapping from S to S_{XY} , the range of the pair (X, Y). (b) Find the probabilities for all values of (X, Y).	**1.5**	**CO1, CO2**
5b	The joint cdf **X**=(*X*,*Y*) for the pair of random variables is given by $F_{X,Y}(x,y) = \begin{cases} 0, & x < 0 \text{ or } y < 0, \\ \frac{xy}{6}, & 0 \le x \le 2,\ 0 \le y \le 3, \\ \frac{x}{2}, & 0 \le x \le 2,\ y > 3, \\ \frac{y}{3}, & x > 2,\ 0 \le y \le 3, \\ 1, & x > 2,\ y > 3. \end{cases}$ Plot the joint cdf and find the marginal cdf of X.	**1.5**	**CO3**

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